

HW2

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Q1

Suppose that $\underline{y}$ is an $n \times 1$ observable random vector that follows Model II.

(a)

If $\underline{\lambda}'_1 \underline{\beta}$ and $\underline{\lambda}'_2 \underline{\beta}$ are both estimable, is $\underline{\lambda}'_1 \underline{\beta} + \underline{\lambda}'_2 \underline{\beta}$ necessarily estimable? Explain your answer.

Answer. Yes.

Note: By definition, $\underline{\lambda}' \underline{\beta}$ is *estimable* if there exists a linear unbiased estimator of it, that is, some $c + \underline{a}' \underline{y}$ with $E(c + \underline{a}' \underline{y}) = \underline{\lambda}' \underline{\beta}$ for all $\underline{\beta}$.

With this definition in hand, also note:

- **Lemma 2.1:** Under Models II–VII, a linear estimator $c + \underline{a}' \underline{y}$ of $\underline{\lambda}' \underline{\beta}$ is unbiased if and only if $c = 0$ and $\underline{\lambda}' = \underline{a}' X$.
- **Theorem 2.1:** Under Models II–VII, $\underline{\lambda}' \underline{\beta}$ is estimable if and only if $\underline{\lambda}' = \underline{a}' X$ for some $n \times 1$ vector $\underline{a}$; said differently, $\underline{\lambda}' \underline{\beta}$ is estimable if and only if $\underline{\lambda}' \in \mathcal{R}(X)$.

Since $\underline{\lambda}'_1 \underline{\beta}$ and $\underline{\lambda}'_2 \underline{\beta}$ are estimable, Theorem 2.1 gives $n \times 1$ vectors $\underline{a}_1$ and $\underline{a}_2$ with $\underline{\lambda}'_1 = \underline{a}'_1 X$ and $\underline{\lambda}'_2 = \underline{a}'_2 X$.

It then follows that $\underline{\lambda}'_1 \underline{\beta} + \underline{\lambda}'_2 \underline{\beta} = (\underline{\lambda}_1 + \underline{\lambda}_2)' \underline{\beta}$ is estimable by Theorem 2.1, as

$$\underline{\lambda}'_1 + \underline{\lambda}'_2 = (\underline{a}_1 + \underline{a}_2)' X \in \mathcal{R}(X).$$

Then, by Lemma 2.1, $(\underline{a}_1 + \underline{a}_2)' \underline{y}$ is a linear unbiased estimator of $(\underline{\lambda}_1 + \underline{\lambda}_2)' \underline{\beta}$.

(b)

If $\underline{\lambda}'_1 \underline{\beta}$ and $\underline{\lambda}'_2 \underline{\beta}$ are both nonestimable, is $\underline{\lambda}'_1 \underline{\beta} + \underline{\lambda}'_2 \underline{\beta}$ necessarily nonestimable? Explain your answer.

Answer. No. We provide proof via counterexample.

Take $p = 2$ and $X = \underline{1}_n \underline{1}'_2$, the $n \times 2$ matrix whose every row is $(1, 1)$, so that $y_i = \beta_1 + \beta_2 + \epsilon_i$.

For any $n \times 1$ vector $\underline{a}$, $\underline{a}' X = (\underline{a}' \underline{1}_n) \underline{1}'_2$, so

$$\mathcal{R}(X) = \{t \underline{1}'_2 : t \in \mathbb{R}\} = \{(t, t) : t \in \mathbb{R}\}.$$

Let $\underline{\lambda}_1 = (1, 0)'$ and $\underline{\lambda}_2 = (0, 1)'$. Then $\underline{\lambda}'_1 = (1, 0) \notin \mathcal{R}(X)$ and $\underline{\lambda}'_2 = (0, 1) \notin \mathcal{R}(X)$, so by Theorem 2.1, $\underline{\lambda}'_1 \underline{\beta} = \beta_1$ and $\underline{\lambda}'_2 \underline{\beta} = \beta_2$ are both nonestimable.

However, $\underline{\lambda}'_1 \underline{\beta} + \underline{\lambda}'_2 \underline{\beta} = \beta_1 + \beta_2$ is estimable by Theorem 2.1, as

$$\underline{\lambda}'_1 + \underline{\lambda}'_2 = (1, 1) \in \mathcal{R}(X).$$

The general statement then is: If $\underline{\lambda}' \underline{\beta}$ is nonestimable, then so is $-\underline{\lambda}' \underline{\beta}$ (as $\underline{\lambda}' \notin \mathcal{R}(X)$ implies $-\underline{\lambda}' \notin \mathcal{R}(X)$), yet $\underline{\lambda}' \underline{\beta} + (-\underline{\lambda}' \underline{\beta}) = 0$ is trivially estimable. So a counterexample exists for any X with $\text{rank}(X) < p$.

(c)

Can $\underline{\lambda}'_1\beta + \underline{\lambda}'_2\beta$ be estimable if $\underline{\lambda}'_1\beta$ is estimable and $\underline{\lambda}'_2\beta$ is nonestimable? Explain your answer.

Answer. No.

Note: By **Corollary 2 to Theorem 2.1**, any linear combination of estimable functions is estimable.

For proof, suppose, for contradiction, that $\underline{\lambda}'_1\beta + \underline{\lambda}'_2\beta$ is estimable. Since $\underline{\lambda}'_1\beta$ is estimable by assumption,

$$\underline{\lambda}'_2\beta = (1)(\underline{\lambda}'_1\beta + \underline{\lambda}'_2\beta) + (-1)(\underline{\lambda}'_1\beta)$$

is a linear combination of estimable functions, and so it is estimable by Corollary 2 to Theorem 2.1.

This contradicts the assumption that $\underline{\lambda}'_2\beta$ is nonestimable, so we must conclude that $\underline{\lambda}'_1\beta + \underline{\lambda}'_2\beta$ is nonestimable.

Q2

Take the model to be Model VII and suppose that

$$\underline{y} = \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix} \quad \text{and} \quad X = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

(a)

Find the orthogonal projections of $\underline{y}$ on $\mathcal{C}(X)$ and on $\mathcal{C}^\perp(X)$.

Answer. The projections are $(2, 3, 0)'$ on $\mathcal{C}(X)$ and $(0, 0, 4)'$ on $\mathcal{C}^\perp(X)$.

Note the following:

- $\mathcal{C}(X) = \{X\underline{b} : \underline{b} \in \mathbb{R}^3\}$ is the span of the columns of X ,
- $\mathcal{C}^\perp(X) = \{\underline{a} \in \mathbb{R}^3 : X'\underline{a} = \underline{0}\}$ is the null space of X' , and
- **Theorem 2.6:** Any $n \times 1$ vector $\underline{y}$ admits a unique decomposition $\underline{y} = \underline{y}_1 + \underline{y}_2$ with $\underline{y}_1 \in \mathcal{C}(X)$ and $\underline{y}_2 \in \mathcal{C}^\perp(X)$. What's more, this decomposition is given by $\underline{y}_1 = X\hat{\underline{\beta}}$ and $\underline{y}_2 = \underline{y} - X\hat{\underline{\beta}}$ for any solution $\hat{\underline{\beta}}$ to the normal equations. And the vectors $\underline{y}_1$ and $\underline{y}_2$ are the orthogonal projections of $\underline{y}$ on $\mathcal{C}(X)$ and $\mathcal{C}^\perp(X)$, resp.

Now, to answer the question at hand:

Finding $\mathcal{C}(X)$. The columns of X are

$$\underline{x}_1 = (1, 2, 0)', \quad \underline{x}_2 = (2, 3, 0)', \quad \underline{x}_3 = (3, 1, 0)'.$$

Each has third coordinate 0, so $\mathcal{C}(X) \subseteq \{(u, v, 0)' : u, v \in \mathbb{R}\}$. For the reverse inclusion, note that

$$-3\underline{x}_1 + 2\underline{x}_2 = (1, 0, 0)' \quad \text{and} \quad 2\underline{x}_1 - \underline{x}_2 = (0, 1, 0)'$$

both lie in $\mathcal{C}(X)$, and every $(u, v, 0)'$ is a linear combination of these two. Hence

$$\mathcal{C}(X) = \{(u, v, 0)' : u, v \in \mathbb{R}\}.$$

Finding $\mathcal{C}^\perp(X)$. For $\underline{a} = (a_1, a_2, a_3)'$,

$$X'\underline{a} = (a_1 + 2a_2, 2a_1 + 3a_2, 3a_1 + a_2)'.$$

Since $\det \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix} = -1 \neq 0$, the first two coordinates vanish only if $a_1 = a_2 = 0$. The third then vanishes as well, and a_3 is unrestricted. Hence

$$\mathcal{C}^\perp(X) = \{(0, 0, w)' : w \in \mathbb{R}\}.$$

The projections. Splitting $\underline{y}$ accordingly,

$$\underline{y} = \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 4 \end{pmatrix},$$

with the first vector in $\mathcal{C}(X)$ and the second in $\mathcal{C}^\perp(X)$. By the uniqueness in Theorem 2.6, these are the orthogonal projections of $\underline{y}$ on $\mathcal{C}(X)$ and $\mathcal{C}^\perp(X)$, resp.

(b)

Determine $\text{rank}(X'X)$.

Answer. $\text{rank}(X'X) = 2$.

Note: By **Lemma 2.3**, $\text{rank}(A'A) = \text{rank}(A)$ for any matrix A .

From part (a), $\mathcal{C}(X)$ is spanned by the two linearly independent vectors $(1, 0, 0)'$ and $(0, 1, 0)'$, so $\text{rank}(X) = \dim \mathcal{C}(X) = 2$. It then follows from Lemma 2.3, with $A = X$, that

$$\text{rank}(X'X) = \text{rank}(X) = 2.$$

(c)

Use the following method and find a generalized inverse of $X'X$.

Let A be an $n \times m$ matrix of rank $r \geq 1$. Let B be a nonsingular $r \times r$ submatrix of A . For matrix A , replace the entries of B by those of $(B^{-1})'$, and replace all other entries of A by 0. Then the transpose of the resulting matrix is a generalized inverse of A .

Answer. We apply the method with $A = X'X$ and B its leading 2×2 submatrix.

Note: By definition, a *generalized inverse* of a matrix A is any matrix G satisfying $AGA = A$.

Here

$$A = X'X = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 3 & 0 \\ 3 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 5 & 8 & 5 \\ 8 & 13 & 9 \\ 5 & 9 & 10 \end{pmatrix},$$

which has rank $r = 2$ by part (b). Then

$$B = \begin{pmatrix} 5 & 8 \\ 8 & 13 \end{pmatrix}, \quad \det(B) = 65 - 64 = 1 \neq 0, \quad (B^{-1})' = B^{-1} = \begin{pmatrix} 13 & -8 \\ -8 & 5 \end{pmatrix}.$$

Replacing the entries of B in A by those of $(B^{-1})'$, and all other entries by 0, gives

$$G = (X'X)^- = \begin{pmatrix} 13 & -8 & 0 \\ -8 & 5 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Side Note: Since B is symmetric, $(B^{-1})' = B^{-1}$; the resulting matrix is then symmetric too, so the final transpose leaves it unchanged.

Check. We have

$$AG = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -7 & 5 & 0 \end{pmatrix} \quad \text{and} \quad AGA = \begin{pmatrix} 5 & 8 & 5 \\ 8 & 13 & 9 \\ 5 & 9 & 10 \end{pmatrix} = A,$$

so G is a generalized inverse of $X'X$.

(d)

Compute a solution $\hat{\underline{\beta}}$ to the normal equations.

Answer. $\hat{\underline{\beta}} = (0, 1, 0)'$.

Note the following:

- **Theorem 2.3:** The normal equations $X'X\hat{\underline{\beta}} = X'y$ are consistent.
- **Theorem 1.2:** A matrix G is a generalized inverse of a matrix A if and only if $\underline{w} = G\underline{c}$ solves $A\underline{w} = \underline{c}$ for every vector $\underline{c}$ for which these equations are consistent.

With these in hand, since the normal equations are consistent (Theorem 2.3), Theorem 1.2 with $A = X'X$ and $\underline{c} = X'y$ gives that $\hat{\underline{\beta}} = (X'X)^- X'y$ solves them. Since

$$X'y = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 3 & 0 \\ 3 & 1 & 0 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 8 \\ 13 \\ 9 \end{pmatrix},$$

taking $(X'X)^- = G$ from part (c) gives

$$\hat{\underline{\beta}} = \begin{pmatrix} 13 & -8 & 0 \\ -8 & 5 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 8 \\ 13 \\ 9 \end{pmatrix} = \begin{pmatrix} 104 - 104 \\ -64 + 65 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

As a direct check, $X'X\hat{\underline{\beta}}$ is the second column of $X'X$, namely $(8, 13, 9)' = X'y$.

(e)

Check that the vectors $X\hat{\underline{\beta}}$ and $\underline{y} - X\hat{\underline{\beta}}$ agree with your results in part (a).

Answer. They agree. From part (d), $X\hat{\underline{\beta}}$ is the second column of X , so

$$X\hat{\underline{\beta}} = \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix}, \quad \underline{y} - X\hat{\underline{\beta}} = \begin{pmatrix} 0 \\ 0 \\ 4 \end{pmatrix},$$

which match the projections in part (a), as Theorem 2.6 requires.

A quick numerical check of parts (c)–(e) in R:

```
X <- matrix(c(1, 2, 3,
              2, 3, 1,
              0, 0, 0), nrow = 3, byrow = TRUE)
y <- c(2, 3, 4)
XtX <- crossprod(X)
G <- matrix(c(13, -8, 0,
              -8, 5, 0,
              0, 0, 0), nrow = 3, byrow = TRUE)

all.equal(XtX %*% G %*% XtX, XtX) # G is a generalized inverse of X'X

## [1] TRUE

bhat <- drop(G %*% crossprod(X, y))
bhat

## [1] 0 1 0
```

```

drop(X %*% bhat)                # projection onto C(X)

## [1] 2 3 0

drop(y - X %*% bhat)            # projection onto C-perp(X)

## [1] 0 0 4

```

Q3

For Model VII, partition $\underline{y}$, X , and $\underline{\epsilon}$ correspondingly as follows:

$$\underline{y} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}, \quad X = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix}, \quad \text{and} \quad \underline{\epsilon} = \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \end{pmatrix}.$$

Suppose that the row spaces of X_1 and X_2 are essentially disjoint, that is, $\mathcal{R}(X_1) \cap \mathcal{R}(X_2) = \{0'\}$. Show that, if $\underline{\lambda}' \in \mathcal{R}(X_1)$, then the least squares estimate of $\underline{\lambda}'\underline{\beta}$ based on the model $\underline{y}_1 = X_1\underline{\beta} + \underline{\epsilon}_1$ is the same as the least squares estimate of $\underline{\lambda}'\underline{\beta}$ based on the model $\underline{y} = X\underline{\beta} + \underline{\epsilon}$.

Hint: Under the full model, the LSE is $\underline{a}'X'\underline{y} = \underline{a}'X'_1\underline{y}_1 + \underline{a}'X'_2\underline{y}_2$ and, under the partial model, the LSE is $\underline{b}'X'_1\underline{y}_1$ where here $X'_1X_1\underline{a} + X'_2X_2\underline{a} = X'X\underline{a} = \underline{\lambda} = X'_1X_1\underline{b}$ (why?); in the last equation, move terms involving X'_2 and X_1 to different sides.

Answer. We provide proof by writing both least squares estimates in terms of solutions to conjugate normal equations, then using the disjointness of $\mathcal{R}(X_1)$ and $\mathcal{R}(X_2)$ to show they coincide.

Note: X_1 is $n_1 \times p$ and X_2 is $n_2 \times p$, with $n_1 + n_2 = n$. The partial model $\underline{y}_1 = X_1\underline{\beta} + \underline{\epsilon}_1$ is itself an instance of Model VII, as $E(\underline{\epsilon}_1) = \underline{0}_{n_1}$ and $\text{Var}(\underline{\epsilon}_1)$ is the leading $n_1 \times n_1$ block of $\text{Var}(\underline{\epsilon})$. So every Model II–VII result below applies to both models.

Side Note: The hint's $\underline{a}$ and $\underline{b}$ are $p \times 1$; we write them as $\underline{r}$ and $\underline{r}_1$ to keep $\underline{a}$ reserved for $n \times 1$ coefficient vectors.

With this in hand, also note the following:

- **Theorem 2.5:** Under Models II–VII, $\underline{\lambda}'\underline{\hat{\beta}} = \underline{\lambda}'\underline{\tilde{\beta}}$ for any two solutions $\underline{\hat{\beta}}$ and $\underline{\tilde{\beta}}$ to the normal equations if and only if $\underline{\lambda}'\underline{\beta}$ is estimable.
- **Theorem 2.7:** Under Models II–VII, $\underline{\lambda}'\underline{\beta}$ is estimable if and only if the conjugate normal equations $X'X\underline{r} = \underline{\lambda}$ are consistent.
- **Theorem 2.8:** Under Models II–VII, the least squares estimate of an estimable $\underline{\lambda}'\underline{\beta}$ equals $\underline{r}'X'\underline{y}$, where $\underline{r}$ is any solution to $X'X\underline{r} = \underline{\lambda}$.
- **Lemma 1.2:** For any matrix M , $\text{tr}(M'M) = 0$ implies $M = 0$.
- **Lemma 1.3(ii):** For any matrices M , P , and Q of appropriate dimensions, $M'MP = M'MQ$ implies $MP = MQ$.

Estimability. Since $\underline{\lambda}' \in \mathcal{R}(X_1)$, we have $\underline{\lambda}' = \underline{t}'X_1$ for some $n_1 \times 1$ vector $\underline{t}$. Hence

$$\underline{\lambda}' = (\underline{t}', \underline{0}'_{n_2}) \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \in \mathcal{R}(X),$$

so $\underline{\lambda}'\underline{\beta}$ is estimable under both models by Theorem 2.1 (see Q1(a)), and by Theorem 2.5 its least squares estimate is well defined under *each* model. The two models have different normal equations, though, so it remains to show that the two estimates coincide.

The two estimates (the hint's “why?”). Since $\underline{\lambda}'\underline{\beta}$ is estimable under both models, Theorem 2.7 gives $p \times 1$ vectors $\underline{r}$ and $\underline{r}_1$ with

$$X'X\underline{r} = \underline{\lambda} \quad \text{and} \quad X'_1X_1\underline{r}_1 = \underline{\lambda},$$

and by Theorem 2.8, the least squares estimates under the full and partial models are $\underline{r}'X'\underline{y}$ and $\underline{r}'_1X'_1\underline{y}_1$, resp. By block multiplication,

$$X'X = X'_1X_1 + X'_2X_2 \quad \text{and} \quad X'\underline{y} = X'_1\underline{y}_1 + X'_2\underline{y}_2,$$

so

$$X'_1X_1\underline{r} + X'_2X_2\underline{r} = \underline{\lambda} = X'_1X_1\underline{r}_1, \quad \text{that is,} \quad X'_1X_1(\underline{r} - \underline{r}_1) = -X'_2X_2\underline{r}.$$

Using *disjointness*. Transposing both sides gives an equality of $1 \times p$ row vectors,

$$\underbrace{[X_1(\underline{r} - \underline{r}_1)]'X_1}_{\in \mathcal{R}(X_1)} = \underbrace{[-X_2\underline{r}]'X_2}_{\in \mathcal{R}(X_2)}.$$

The left side is $\underline{u}'X_1$ with $\underline{u} = X_1(\underline{r} - \underline{r}_1) \in \mathbb{R}^{n_1}$, so it lies in $\mathcal{R}(X_1)$; similarly, the right side lies in $\mathcal{R}(X_2)$. Since $\mathcal{R}(X_1) \cap \mathcal{R}(X_2) = \{\underline{0}'\}$, both sides equal $\underline{0}'_p$. This gives two facts:

1. $\underline{r}'X'_2X_2 = \underline{0}'_p$, so $(X_2\underline{r})'(X_2\underline{r}) = \underline{r}'X'_2X_2\underline{r} = 0$. Then, by Lemma 1.2 with $M = X_2\underline{r}$ (an $n_2 \times 1$ matrix, for which $\text{tr}(M'M) = M'M = 0$), $X_2\underline{r} = \underline{0}_{n_2}$.
2. $X'_1X_1\underline{r} = X'_1X_1\underline{r}_1$. Then, by Lemma 1.3(ii) with $M = X_1$, $P = \underline{r}$, and $Q = \underline{r}_1$, $X_1\underline{r} = X_1\underline{r}_1$.

Conclusion. Combining these,

$$\underline{r}'X'\underline{y} = \underline{r}'X'_1\underline{y}_1 + \underline{r}'X'_2\underline{y}_2 = (X_1\underline{r})'\underline{y}_1 + (X_2\underline{r})'\underline{y}_2 = (X_1\underline{r}_1)'\underline{y}_1 + 0 = \underline{r}'_1X'_1\underline{y}_1,$$

so the least squares estimates of $\underline{\lambda}'\underline{\beta}$ under the two models coincide. Said differently, since $X_2\underline{r} = \underline{0}_{n_2}$, the observations in $\underline{y}_2$ get zero weight in the full-model estimate of $\underline{\lambda}'\underline{\beta}$. $\square$

Q4

Show that, under Model II, $\underline{a}'\underline{y}$ is a BLUE of its expectation if and only if $\underline{a} \in \mathcal{C}(X)$.

Answer. We provide proof by splitting $\underline{a} = P_X\underline{a} + (I_n - P_X)\underline{a}$, where $P_X = X(X'X)^{-}X'$, and showing that $\underline{a}'\underline{y}$ attains the BLUE's variance exactly when the second piece is zero.

Note: $\underline{a}$ is $n \times 1$, and under Model II, $E(\underline{\epsilon}) = \underline{0}_n$ and $\text{Var}(\underline{y}) = \text{Var}(\underline{\epsilon}) = \sigma^2 I_n$ with $\sigma^2 > 0$.

Also note the following:

- **Lemma 1.4:** For any matrix X , $X'X(X'X)^{-}X' = X'$.
- **Corollary 1(ii) to Lemma 1.4:** For any matrix X , $X(X'X)^{-}X'X = X$.
- **Corollary 3 to Lemma 1.5:** For any matrix X , $X(X'X)^{-}X'$ is symmetric.
- **Corollary to Lemma 2.4:** For a random vector $\underline{x}$ with $\text{Var}(\underline{x}) = \Sigma$ and a matrix of constants L , $\text{Var}(L'\underline{x}) = L'\Sigma L$.
- **Theorem 2.9 (Gauss–Markov):** Under Models II–VII, if $\underline{\lambda}'\underline{\beta}$ is estimable and $\widehat{\underline{\beta}}$ is any solution to the normal equations, then $\underline{\lambda}'\widehat{\underline{\beta}}$ is a linear unbiased estimator of $\underline{\lambda}'\underline{\beta}$. What's more, under Model II, the variance of $\underline{\lambda}'\widehat{\underline{\beta}}$ is uniformly less than that of any other linear unbiased estimator of $\underline{\lambda}'\underline{\beta}$; said differently, $\underline{\lambda}'\widehat{\underline{\beta}}$ is the BLUE.

Properties of P_X . The $n \times n$ matrix P_X satisfies

- $P_X X = X$, by Corollary 1(ii) to Lemma 1.4,
- $P'_X = P_X$, by Corollary 3 to Lemma 1.5, and
- $P_X^2 = X(X'X)^{-}[X'X(X'X)^{-}X'] = X(X'X)^{-}X' = P_X$, by Lemma 1.4.

It then follows that $P'_X(I_n - P_X) = P_X - P_X^2 = \underline{0}_{n \times n}$.

The BLUE of $E(\underline{a}'\underline{y})$. We have $E(\underline{a}'\underline{y}) = \underline{a}'X\underline{\beta} = \underline{\lambda}'\underline{\beta}$ with $\underline{\lambda}' = \underline{a}'X \in \mathcal{R}(X)$. By Lemma 2.1 (see Q1(a)), $\underline{a}'\underline{y}$ is a linear unbiased estimator of $\underline{\lambda}'\underline{\beta}$, and so $\underline{\lambda}'\underline{\beta}$ is estimable. By Theorems 2.3 and 1.2 (see Q2(d)), $\widehat{\underline{\beta}} = (X'X)^{-}X'\underline{y}$ solves the normal equations, so by Theorem 2.9, the BLUE of $\underline{\lambda}'\underline{\beta}$ is

$$\underline{\lambda}'\widehat{\underline{\beta}} = \underline{a}'X(X'X)^{-}X'\underline{y} = \underline{a}'P_X\underline{y} = (P_X\underline{a})'\underline{y},$$

where the last step uses $P_X' = P_X$.

Comparing variances. Applying the Corollary to Lemma 2.4 with $L = \underline{a}$ and $\Sigma = \sigma^2 I_n$, and writing $\underline{a} = P_X\underline{a} + (I_n - P_X)\underline{a}$,

$$\text{Var}(\underline{a}'\underline{y}) = \sigma^2 \underline{a}'\underline{a} = \sigma^2 (P_X\underline{a})'(P_X\underline{a}) + 2\sigma^2 \underline{a}'P_X'(I_n - P_X)\underline{a} + \sigma^2 [(I_n - P_X)\underline{a}]'[(I_n - P_X)\underline{a}].$$

The middle term vanishes because $P_X'(I_n - P_X) = 0_{n \times n}$, and the first term is $\text{Var}((P_X\underline{a})'\underline{y})$, again by the Corollary to Lemma 2.4. Hence

$$\text{Var}(\underline{a}'\underline{y}) = \text{Var}((P_X\underline{a})'\underline{y}) + \sigma^2 [(I_n - P_X)\underline{a}]'[(I_n - P_X)\underline{a}]. \quad (\text{Q4.1})$$

When $\underline{a}'\underline{y}$ is a BLUE. Since $\underline{a}'\underline{y}$ is linear and unbiased for $\underline{\lambda}'\underline{\beta}$, and $(P_X\underline{a})'\underline{y}$ is the BLUE, $\underline{a}'\underline{y}$ is a BLUE if and only if $\text{Var}(\underline{a}'\underline{y}) = \text{Var}((P_X\underline{a})'\underline{y})$. Because $\sigma^2 > 0$, (Q4.1) shows this holds if and only if $[(I_n - P_X)\underline{a}]'[(I_n - P_X)\underline{a}] = 0$. By Lemma 1.2 (see Q3), applied to the $n \times 1$ matrix $(I_n - P_X)\underline{a}$, this holds if and only if $(I_n - P_X)\underline{a} = \underline{0}_n$, that is, $\underline{a} = P_X\underline{a}$.

When $\underline{a} = P_X\underline{a}$. Recall that $\underline{a} \in \mathcal{C}(X)$ if and only if $\underline{a} = X\underline{b}$ for some $p \times 1$ vector $\underline{b}$.

- ($\Rightarrow$) If $\underline{a} = P_X\underline{a}$, then $\underline{a} = X\underline{b}$ with $\underline{b} = (X'X)^{-}X'\underline{a}$, so $\underline{a} \in \mathcal{C}(X)$.
- ($\Leftarrow$) If $\underline{a} = X\underline{b}$, then $P_X\underline{a} = P_XX\underline{b} = X\underline{b} = \underline{a}$, since $P_XX = X$.

Hence $\underline{a}'\underline{y}$ is a BLUE of its expectation if and only if $\underline{a} \in \mathcal{C}(X)$. □

Q5

(a)

Show that, under Model II, the problem of determining the BLUE $\underline{a}'\underline{y}$ of an estimable function $\underline{\lambda}'\underline{\beta}$ is equivalent to that of minimizing $\underline{a}'\underline{a}$ subject to the constraints $\underline{a}'X = \underline{\lambda}'$.

Answer. The two problems share a feasible set, and their objectives differ only by the constant factor $\sigma^2 > 0$.

Note: $\underline{a}$ is $n \times 1$ and $\underline{\lambda}$ is $p \times 1$.

By Lemma 2.1 (see Q1(a)), a linear estimator $c + \underline{a}'\underline{y}$ is unbiased for $\underline{\lambda}'\underline{\beta}$ if and only if $c = 0$ and $\underline{a}'X = \underline{\lambda}'$. So the linear unbiased estimators of $\underline{\lambda}'\underline{\beta}$ are exactly

$$\{\underline{a}'\underline{y} : \underline{a}'X = \underline{\lambda}'\},$$

which is nonempty because $\underline{\lambda}'\underline{\beta}$ is estimable. Under Model II, $\text{Var}(\underline{y}) = \sigma^2 I_n$, so by the Corollary to Lemma 2.4 (see Q4), each has variance

$$\text{Var}(\underline{a}'\underline{y}) = \underline{a}'(\sigma^2 I_n)\underline{a} = \sigma^2 \underline{a}'\underline{a}.$$

Since $\sigma^2 > 0$ does not depend on $\underline{a}$, it then follows that $\underline{a}$ minimizes $\text{Var}(\underline{a}'\underline{y})$ over this class if and only if it minimizes $\underline{a}'\underline{a}$ subject to $\underline{a}'X = \underline{\lambda}'$ (equivalently, $X'\underline{a} = \underline{\lambda}$). And since the minimizing $\underline{a}$ does not depend on σ^2 , the resulting estimator has minimum variance uniformly in σ^2 . □

(b)

The Lagrangian function associated with this minimization problem is

$$L(\underline{a}, \underline{\rho}) = \underline{a}'\underline{a} + 2\underline{\rho}'(X'\underline{a} - \underline{\lambda}).$$

Use the vector calculus to differentiate the Lagrangian function with respect to $\underline{a}$ and the Lagrangian multipliers, and set these derivatives equal to zero.

Answer.

Note: There is one multiplier per constraint in $X'\underline{a} = \underline{\lambda}$, so $\underline{\rho}$ is $p \times 1$.

Also note the following identities from Section 2.3 of the notes, for a vector variable $\underline{\beta}$:

- For any constant vector $\underline{h}$, $\partial(\underline{h}'\underline{\beta})/\partial\underline{\beta} = \underline{h}$.
- For any constant matrix X , $\partial(\underline{\beta}'X'X\underline{\beta})/\partial\underline{\beta} = 2X'X\underline{\beta}$.

Applying these term by term:

- *With respect to $\underline{a}$.* The second identity, with I_n in place of X and $\underline{a}$ in place of $\underline{\beta}$, gives $\partial(\underline{a}'\underline{a})/\partial\underline{a} = 2\underline{a}$. Writing $\underline{\rho}'X'\underline{a} = (X\underline{\rho})'\underline{a}$, the first identity with $\underline{h} = X\underline{\rho}$ gives $\partial(\underline{\rho}'X'\underline{a})/\partial\underline{a} = X\underline{\rho}$. The term $-2\underline{\rho}'\underline{\lambda}$ does not involve $\underline{a}$.
- *With respect to $\underline{\rho}$.* The term $\underline{a}'\underline{a}$ does not involve $\underline{\rho}$. Writing $2\underline{\rho}'(X'\underline{a} - \underline{\lambda}) = 2(X'\underline{a} - \underline{\lambda})'\underline{\rho}$, the first identity with $\underline{h} = X'\underline{a} - \underline{\lambda}$ applies.

Together, the $n \times 1$ and $p \times 1$ gradients are

$$\frac{\partial L}{\partial \underline{a}} = 2\underline{a} + 2X\underline{\rho}, \quad \frac{\partial L}{\partial \underline{\rho}} = 2(X'\underline{a} - \underline{\lambda}).$$

Setting these equal to zero gives the system

$$\underline{a} + X\underline{\rho} = \underline{0}_n, \quad X'\underline{a} = \underline{\lambda}. \quad (\text{Q5.1})$$

(c)

Show that the resulting equations have a solution if and only if the conjugate normal equations are consistent.

Answer. The link between the two systems is the substitution $\underline{r} = -\underline{\rho}$, with $\underline{a} = X\underline{r}$.

($\Rightarrow$) Suppose $(\underline{a}_0, \underline{\rho}_0)$ solves (Q5.1). The first equation gives $\underline{a}_0 = -X\underline{\rho}_0$, and substituting into the second,

$$\underline{\lambda} = X'\underline{a}_0 = X'X(-\underline{\rho}_0).$$

So $\underline{r} = -\underline{\rho}_0$ solves $X'X\underline{r} = \underline{\lambda}$, and the conjugate normal equations are consistent.

($\Leftarrow$) Suppose $X'X\underline{r}_0 = \underline{\lambda}$ for some $p \times 1$ vector $\underline{r}_0$. Set $\underline{a}_0 = X\underline{r}_0$ and $\underline{\rho}_0 = -\underline{r}_0$. Then

$$\underline{a}_0 + X\underline{\rho}_0 = X\underline{r}_0 - X\underline{r}_0 = \underline{0}_n \quad \text{and} \quad X'\underline{a}_0 = X'X\underline{r}_0 = \underline{\lambda},$$

so $(\underline{a}_0, \underline{\rho}_0)$ solves (Q5.1). □

(d)

Solve your equations for the minimizing value of $\underline{a}$.

Answer. $\underline{a}_* = X(X'X)^{-1}\underline{\lambda}$.

Note the following:

- **Theorem 1.3:** All solutions to a consistent matrix equation $AW = Y$ are generated by $W = A^-Y + (I - A^-A)Z$ as Z ranges over all matrices of appropriate dimensions. (We write W for the notes' unknown to avoid a clash with the design matrix X .)

- **Corollary to Lemma 1.1:** For any matrix X , $[(X'X)^-]'$ is a generalized inverse of $X'X$.

A solution exists. Since $\underline{\lambda}'\underline{\beta}$ is estimable, the conjugate normal equations are consistent by Theorem 2.7 (see Q3), so (Q5.1) has a solution by part (c).

All solutions. The first equation of (Q5.1) gives $\underline{a} = -X\underline{\rho}$. Substituting into the second gives $X'X(-\underline{\rho}) = \underline{\lambda}$, a consistent system in the $p \times 1$ unknown $-\underline{\rho}$. Applying Theorem 1.3 with $A = X'X$, $W = -\underline{\rho}$, and $\underline{Y} = \underline{\lambda}$, all solutions are

$$-\underline{\rho} = (X'X)^-\underline{\lambda} + [I_p - (X'X)^-X'X]\underline{z}, \quad \underline{z} \in \mathbb{R}^p.$$

Hence

$$\underline{a} = -X\underline{\rho} = X(X'X)^-\underline{\lambda} + [X - X(X'X)^-X'X]\underline{z} = X(X'X)^-\underline{\lambda},$$

since $X(X'X)^-X'X = X$ by Corollary 1(ii) to Lemma 1.4 (see Q4). So every solution of (Q5.1) has the same $\underline{a}$ -component,

$$\underline{a}_* = X(X'X)^-\underline{\lambda}.$$

The stationary point is the minimum. Equations (Q5.1) are necessary conditions only, so we verify directly that $\underline{a}_*$ minimizes $\underline{a}'\underline{a}$ subject to $X'\underline{a} = \underline{\lambda}$. Let $\underline{a}$ be any $n \times 1$ vector with $X'\underline{a} = \underline{\lambda}$, and set $\underline{d} = \underline{a} - \underline{a}_*$, so that $X'\underline{d} = \underline{\lambda} - \underline{\lambda} = \underline{0}_p$. Since $\underline{a}_* = X\underline{r}_*$ with $\underline{r}_* = (X'X)^-\underline{\lambda}$,

$$\underline{a}'_*\underline{d} = \underline{r}'_*X'\underline{d} = 0,$$

and therefore

$$\underline{a}'\underline{a} = (\underline{a}_* + \underline{d})'(\underline{a}_* + \underline{d}) = \underline{a}'_*\underline{a}_* + \underline{d}'\underline{d} \geq \underline{a}'_*\underline{a}_*.$$

Equality holds if and only if $\underline{d}'\underline{d} = 0$, that is, $\underline{d} = \underline{0}_n$ (by Lemma 1.2, see Q3). So $\underline{a}_*$ is the unique minimizer.

Side Note: Uniqueness also means $X(X'X)^-\underline{\lambda}$ does not depend on the choice of generalized inverse.

The resulting BLUE. By part (a), the BLUE is

$$\underline{a}'_*\underline{y} = \underline{\lambda}'[(X'X)^-]'X'\underline{y}.$$

By the Corollary to Lemma 1.1, $[(X'X)^-]'$ is itself a generalized inverse of $X'X$, so by Theorems 2.3 and 1.2 (see Q2(d)), $\widehat{\underline{\beta}} = [(X'X)^-]'X'\underline{y}$ solves the normal equations, and

$$\underline{a}'_*\underline{y} = \underline{\lambda}'\widehat{\underline{\beta}}.$$

That is, the BLUE of $\underline{\lambda}'\underline{\beta}$ is its least squares estimator, in agreement with Theorem 2.9 (see Q4). $\square$